

```
#include <stdio.h>
```

```
int mutex = 1, full = 0, empty = 1, x = 0;
```

```
void producer() {
```

```
    mutex--;
```

```
    empty--;
```

```
    full++;
```

```
    x++;
```

```
    printf("Producer produces item %d\n", x);
```

```
    mutex++;
```

```
}
```

```
void consumer() {
```

```
    mutex--;
```

```
    full--;
```

```
    empty++;
```

```
    printf("Consumer consumes item %d\n", x);
```

```
    x--;
```

```
    mutex++;
```

```
}
```

```
int main() {
```

```
    int ch;
```

```
    while(1) {
```

```
printf("\n1. Producer\n2. Consumer\n3. Exit\n");  
printf("Enter your choice: ");  
scanf("%d", &ch);  
  
switch(ch) {  
    case 1:  
        if((mutex==1)&&(empty!=0))  
            producer();  
        else  
            printf("Buffer is Full!\n");  
        break;  
  
    case 2:  
        if((mutex==1)&&(full!=0))  
            consumer();  
        else  
            printf("Buffer is Empty!\n");  
        break;  
  
    case 3:  
        return 0;  
    }  
}  
}
```